Bioresource Technology 427 (2025) 132398

Contents lists available at ScienceDirect

Bioresource Technology

journal homepage: www.elsevier.com/locate/biortech

In-situ intermittent micro-aeration in food waste and sewage sludge anaerobic co-digestion: Performance, stability, and microbial dynamics☆

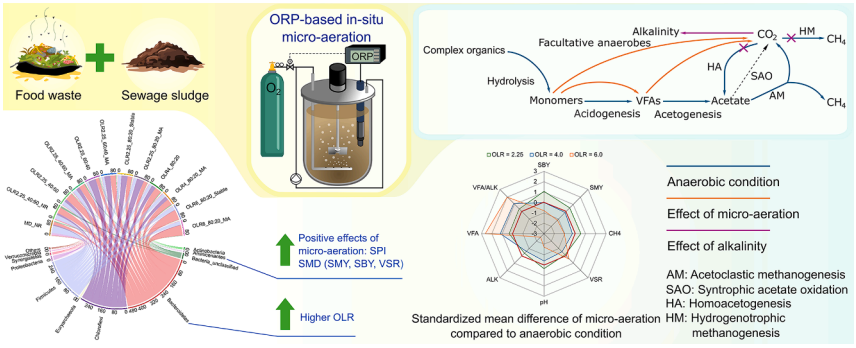
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HIGHLIGHTS

- Effective mixing is key to enhancing oxygen mass transfer in in-situ micro-aeration.
- Homoacetogenesis outcompetes hydrogenotrophic methanogenesis during micro-aeration.
- Micro-aeration enhances performance and stability when alkalinity <5000 mg$CaCO_3$/L.
- Higher alkalinity resulted in lower hydrogenotrophic methanogenesis.
- Micro-aeration enhanced fermentation of complex substrates by facultative anaerobes.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Organic waste
Resource recovery
Oxidation-reduction potential
Homoacetogenesis
Aminicenantes
Facultative anaerobes

ABSTRACT

Anaerobic co-digestion (AcoD) of sewage sludge (SS) with food waste (FW) is a practical approach in urban areas due to spatio-temporal availability of these co-substrates. While micro-aeration could enhance hydrolysis and control acidification during mono-digestion, the effects of micro-aeration on AcoD remains unclear. This study explored the influence of oxidation–reduction potential (ORP)-based in-situ micro-aeration on AcoD performance of SS and FW. Although mono-digestion of FW failed due to acidification and micro-aeration of SS was unstable due to high solids content and viscosity, micro-aeration improved specific biogas and methane yields by 21.1 % and 13.1 %, respectively, at the FW:SS ratio of 60:40 ((volatile solids (VS) basis). This improvement was observed at an organic loading rate of 2.25 g VS/L·d and alkalinity below 5000 mg $CaCO_3$/L. The observed positive effect of micro-aeration was accompanied by changes in relative abundance of Aminicenantes. Functional analysis suggested that AcoD was driven by homoacetogenesis and acetoclastic methanogenesis.

☆ This article is part of a special issue entitled: 'IWA AD18 (2024)' published in Bioresource Technology.

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https://doi.org/10.1016/j.biortech.2025.132398
Received 30 November 2024; Received in revised form 12 March 2025; Accepted 12 March 2025
Available online 15 March 2025
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1. Introduction

The rapid population growth and urbanization has further aggravated issues with organic waste management, especially food waste (FW) and sewage sludge (SS). FW accounts for 62 % of household waste by mass, amounting to 77 kg per capita annually (UNEP, 2024). In 2022, total of 1.05 billion metric tons of FW were estimated to be generated from household, food service, and retailing (UNEP, 2024). Despite efforts to address the issue, no significant improvement has been made in food loss, which remained at 13 % of the food produced (FAO, 2023). This amount of FW accounts for significant land and carbon footprints since food loss and waste generate an estimated 8–10 % of global greenhouse gases (GHGs) and take up nearly 30 % of agricultural land (UNEP, 2024). In the U.S. alone, food loss and waste management correspond to 170 million metric tons (MMT) of CO_2e GHG emissions (excluding landfills). SS, a byproduct of activated sludge systems in wastewater treatment, is generated at a rate of 45 dry MMT per year (Gao et al., 2020). SS could contain organic compounds, trace elements, and organic nitrogen, though the concern is the potential presence of heavy metals. Among treatment methods like composting and incineration, anaerobic digestion (AD) stands out as the method with the lowest GHG emissions because it produces bioenergy (Piippo et al., 2018).

Anaerobic co-digestion (AcoD) has emerged as a promising and practical approach for managing organic wastes, offering improved system performance and stability due to the complementary characteristics of different substrates, the enrichment of microbial consortia, improved quality of digestate, and reduced GHG emissions (Chuenchart et al., 2024). Recently, significant efforts have been made to improve biodegradability and enhance microbial synergy in AD process. One such approach, in-situ micro-aeration, involves introducing a small amount of oxygen into the aqueous phase of the digester at dosing rates of 0.01–5.00 mL-O_2/L·d. However, detecting low oxygen levels in AD systems is challenging due to the limitations of DO probes (> 0.1 mg-O_2/L), which are not designed for high-solids environments (Yu et al., 2020). As oxygen input increases, the system becomes more oxidative. Oxidation-reduction potential (ORP) serves as an indirect indicator of oxygen levels in AD systems and helps maintain an optimal range for effective micro-aeration (Nghiem et al., 2014; Nguyen et al., 2019).

Due to its high electronegativity, oxygen serves as a rapid electron acceptor within AD systems, facilitating organic matter degradation by facultative anaerobes, volatile fatty acids (VFAs) oxidation through β-oxidation, and sulfide removal via desulfurization (Nguyen et al., 2019). Therefore, micro-aeration can either improve AD system performance and stability or lead to acidification, depending on factors such as micro-aeration intensity, substrate composition, digester operational conditions, and microbial niches (Fu et al., 2023; Li et al., 2024). For example, in SS mono-digestion, micro-aeration at 12.5 mL-air/L·d improved specific biogas yield (SBY) and volatile solids (VS) reduction by 16 % and 18 %, respectively (Zhou et al., 2021). Similarly, in FW mono-digestion, micro-aeration within acidogenic reactor at 258 L-air/kg total solids (TS)/d improved acetic acid production by 3-folds and enhanced soluble chemical oxygen demand (sCOD) production, as well as carbohydrate and protein degradation, by 21 %, 22 %, and 38 %, respectively. This resulted in 17 % improvement in specific methane yield (SMY) compared to non-micro-aerated condition. However, the same study reported that micro-aeration at intensities either higher or lower than 258 L-air/kg TS/d resulted in a 17–26 % reduction in SMY (Xu et al., 2014).

Despite the growing interest in the application of micro-aeration technology in AD field, studies on its effects in AcoD systems remain limited. For example, micro-aeration pretreatment (4 days, 37.5 mL-O_2/L_R·d) improved sCOD of hydrolysate by 13 %, leading to 23 % increase in SMY compared to non-micro-aerated condition in AcoD of FW and brown water (150 g: 2 L) (Lim and Wang, 2013). In a sequencing batch system, in-situ micro-aeration (375 mL-air/L·d) improved SMY by 22 %, as well as effluent sCOD by 67 % and VFAs by 58 %, in the AcoD of poultry litter and wheat straw (1:1, mass basis) compared to non-micro-aerated condition (Zhan et al., 2023). In contrast, in-situ micro-aeration (1000 mL-air/L·d) during AcoD of swine manure with corn silage (17.1:111.2, mass basis) resulted in higher acidification, increasing VFAs and lactic acid concentrations by 20 % and 19 %, respectively, compared to non-micro-aerated condition (Cao et al., 2022). Given these mixed and inconclusive effects, micro-aeration may help stabilize AcoD system for FW and SS co-digestion by regulating VFA accumulation across different co-substrate ratios. Furthermore, no long-term studies have explored the effects of in-situ micro-aeration on AcoD system performance and microbial community dynamics.

This study evaluated the effect of ORP-based in-situ micro-aeration on the performance and stability of an AcoD system, compared to non-micro-aerated (anaerobic) conditions, using varying co-substrate ratios and organic loading rates (OLRs). Initially, the FW to SS ratio (FW:SS, VS basis) was optimized, with the highest SMY observed at FW:SS ratio of 80:20. Subsequently, the OLR was gradually increased from 2.25 to 4.0 and 6.0 g VS/L·d in a stepwise approach. Additionally, microbial community dynamics were monitored throughout the long-term semi-continuous operation of the AcoD system.

2. Materials and methods

2.1. Substrate and inoculum

Post-consumer FW, which was a mixture of cooked food, fruit peel, and vegetable scraps, was collected for a week in the spring of 2020 from Gateway Café at University of Hawai'i at Mānoa. The FW mixture was blended using a mechanical blender (Vitamix VM0185A, USA), sieved through a 5-mm pore size sieve, and thoroughly mixed to ensure homogenization. SS, which was a mixture of primary solids and waste activated sludge retrieved from the belt filter press, and anaerobically digested waste activated sludge (ADWAS) were obtained in the spring of 2020 from Hawai'i Kai wastewater treatment plant, Honolulu, HI, USA. Both FW and SS were stored at −20 °C, while the (ADWAS) was stored at 4 °C until further use. The characteristics of FW, SS, and ADWAS used in this study are summarized in Karki et al. (2022). ADWAS was used as an inoculum at the initial stage of digester operation. After the system was started, the digestate from each stage of digester operation was collected and stored in an air-tight container at 25 °C.

2.2. Semi-continuous system operation and in-situ micro-aeration

AD study was conducted using two, 8-L continuous stirred-tank reactors (CSTRs) with working volume of 6 L and mixing speed of 120 rpm, and operated in semi-continuous mode under mesophilic condition (35 ± 1 °C) at a hydraulic retention time (HRT) of 20 days. Each reactor was installed with an ORP control system (pH/ORP 400, Cole-Parmer, USA) as previously described in Nguyen et al. (2019) and a perforated aeration tube for ORP-based in-situ micro-aeration. The mono-digestion of SS and FW was evaluated in parallel, starting at an OLR of 1.00 g VS/L·d. The FW mono-digestion failed even at an OLR of 1.00 g VS/L·d due to acidification (pH = 6.80 and total volatile fatty acid-to-total alkalinity (VFA/ALK) ratio = 0.530 g HAc (acetic acid equivalent)/g $CaCO_3$) and was discontinued. The SS mono-digestion was tested at an OLR up to 2.25 g VS/L·d —the maximum possible at an HRT of 20 days due to intrinsic characteristics of SS. AcoD of FW and SS was initiated at the FW:SS ratio of 40:60 (VS basis) at an OLR of 2.25 g VS/L·d. ORP-based in-situ micro-aeration was initially tested at +10 mV above the baseline ORP of −569 mV. A significant amount of injected oxygen escaped into the headspace. To address this issue and enhance oxygen mass transfer, biogas was recirculated through the bottom of the reactor. Then the effect of intermittent (every alternate day) in-situ micro-aeration at an ORP of +25 mV above the baseline ORP (under anaerobic conditions) was evaluated at an OLR of 2.25 g VS/L·d across different co-substrate mixing ratios, with increasing content of FW in the mixture (i.e., FW:

SS ratios of 40:60, 60:40, and 80:20). The effect of in-situ micro-aeration on AcoD of FW and SS was further studied at OLRs of 4.0 and 6.0 g VS/L·d, using the FW:SS ratio of 80:20, which resulted in the highest SMY. The average oxygen injection varied from 29 to 82 mL-O_2/L·d, depending on the intrinsic oxygen needed to achieve the targeted ORP. Each stage was maintained for at least 20 days in a quasi-steady state, defined by no significant difference ($p > 0.05$) in SMY for five consecutive days of operation (see supplementary materials). Part of the data from non-microaerated condition was used for machine learning model verification in our previous study (Chuenchart et al., 2024).

2.3. Analytical methods

Biogas production and composition, pH, VFA/ALK ratio, and individual VFAs were analyzed periodically. The volume of the biogas was measured using a drum-type gas counter (TG 0.5/5 drum-type gas meter, Ritter, Germany) and normalized to standard temperature (273 K) and pressure (1 atm). The composition of biogas was determined using a gas chromatograph equipped with a thermal conductivity detector (GC-TCD) (490 Micro GC System, Agilent, USA). pH, alkalinity (ALK), total VFA, and VFA/ALK ratio were measured using an autotitrator (TitraLab AT1000, Hach, USA) with a pH probe (PHC805). TS and VS were determined following the Standard Methods (APHA, 2017). Individual VFA concentrations were determined using a gas chromatograph with a flame ionization detector (GC-FID) (GC-2014, Shimadzu, Japan). sCOD and total ammonia nitrogen were determined using HACH TNT 823 and TNT 834, respectively.

2.4. Microbial community analysis

Biomass samples were collected from the digesters across all experimental conditions and were stored at −20 °C until further processed. DNA extractions were performed in triplicates using the Maxwell 16 RSC Blood DNA kit (Promega, USA), according to modified manufacturer's instructions. Briefly, the samples were initially mixed with 0.5 g zirconium beads and 600 uL lysis buffer. Three cycles of 2 min bead beating using the Mini-BeadBeater-24 (BioSpec Products, USA) and centrifugation at 12,500×*g* for 10 min were performed to collect the supernatant, after which proteinase K was added. After incubation at 56 °C for 20 min, DNA extracts were obtained following automated extraction using the Maxwell® RSC 48 Instrument (Promega, USA). DNA concentrations and quality were determined using NanoDrop One Spectrophotometer (Thermo Fisher, USA). Microbial sequencing was performed at the University of Michigan's Microbiome Core facility (University of Michigan, Ann Arbor, MI, USA), targeting the V4 region of the 16S rRNA gene, using modified earth microbiome project (EMP) primers (Kozich et al., 2013). Multiplexed amplicons were sequenced using Illumina MiSeq using the MiSeq reagent kit V2 (2 × 250 bp reads) according to the manufacturer's instructions with modifications found in the EMP SOP (Caporaso et al., 2012; Parada et al., 2016). FASTQ files were generated for paired-end reads and were analyzed using the Schloss MiSeq SOP on the Mothur bioinformatics platform. Sequences were filtered, and chimeras were removed using the UCHIME algorithm, after which the sequences were aligned using the SILVA reference database (Quast et al., 2013). Further analysis was carried out using operational taxonomic unit (OTU) clustering based on an average neighbor algorithm at a 3 % cutoff limit. The software package Tax4Fun2 was used in the R environment to predict functional capabilities using the taxonomic profile to predict the metabolic pathways in the Kyoto Encyclopedia of Genes and Genomes (KEGG). OTUs were mapped to reference sequences, and the taxonomic abundance was converted into the relative abundance of enzyme-encoding genes associated with KEGG pathways and orthologs. All raw sequences are available in the National Center for Biotechnology Information (NCBI) Sequence Read Archive (SRA) under BioProject ID: PRJNA1217127.

2.5. Statistical analysis

The effect of co-substrate mixing ratios, OLRs, and micro-aeration on SBY, SMY, VS reduction, pH, ALK, VFA, and VFA/ALK ratio were analyzed using *t*-test with a significance level of $p < 0.05$. Violin plot and statistical analysis were conducted using the following packages: car, ggplot2, Imer4, meta and ImerTest in Rstudio (R studio v.1.2.5033, USA). For the microbial community analysis, the redundancy analysis (RDA) was applied, the α-diversity and β-diversity were analyzed, and statistical significance was calculated following two-tailed unpaired Student's *t*-test, analysis of molecular variance (AMOVA), and homogeneity of molecular variance (HOMOVA) with statistical significance at $p < 0.05$. Summary statistical analyses were conducted in exploratory and comparative manners, including Pearson correlation analysis, RDA, and Hedges' g standardized mean differences (SMD, Eq. (1) with a random effect model. To account for the effects of micro-aeration, a standardized performance index (SPI) was used. The SPI was calculated as the average of standardized mean differences (effect sizes) of performance parameters (x) including SMY, SBY, and VS reduction (Eq. (2).

$$\mathrm{SMD}_x = \frac{\overline{x}_{AN} - \overline{x}_{MA}}{SD^{*}_{pooled}} \quad (1)$$

$$\mathrm{SPI} = \overline{SMD}_p = \frac{SMD_{SMY} + SMD_{SBY} + SMD_{VSR}}{N_p} \quad (2)$$

where, SMD_x: Standardized mean different of parameter x; $\overline{x}_{AN}$ and $\overline{x}_{MA}$: Means of performance parameters of anaerobic and micro-aerobic conditions, respectively; SD^{*}_{pooled}: Pooled and weighted standard deviation; N_p: Number of parameters.

3. Results and discussion

3.1. Effects of co-substrate ratio and micro-aeration on system performance and stability

3.1.1. Start-up phases

At the beginning of SS mono-digestion (OLR = 1.00 g VS/L·d), SMY reached 210 NmL/g VS within the first week (Fig. 1**a**) with the VFA/ALK ratio of 0.198 g HAc/g $CaCO_3$ and pH of 7.17. During this stage (i.e., SS mono-digestion at an OLR of 1.00 g VS/L·d), SMY varied between 170 and 302 NmL/g VS with the VFA/ALK ratio of 0.207 ± 0.057 g HAc/g $CaCO_3$ (pH = 7.27 ± 0.09). Subsequently, OLR was increased to 2.25 g VS/L·d (Stage I) (Fig. 1**a**). At this stage, SBY and SMY were 260 ± 29 and 190 ± 19 NmL/g VS, respectively, with VS reduction of 38.8 ± 2.3 % (Fig. 1**b, c, d**). pH remained within an acceptable range at 7.78 ± 0.05 with the VFA/ALK ratio of 0.299 ± 0.040 g HAc/g $CaCO_3$ (Fig. 1**e, f**), indicating stable digester performance. However, micro-aeration (+25 mV) during the mono-digestion of SS at an OLR of 2.25 g VS/L·d (days 178–188) led to digester instability, resulting in near failure with SMY dropping to 89 NmL/g VS. This instability may have been caused by the high viscosity of the digester contents (TS > 4.0 %), which hindered oxygen mass transfer. A significant portion of the injected oxygen escaped to the headspace, resulting in 10 % oxygen content (v/v) in the biogas.

To address the high viscosity issue of the digester contents, AcoD with FW was initiated at the SS:FW ratio of 40:60 (VS basis) while maintaining the same OLR of 2.25 g VS/L·d (Stage II). During this stage, SBY and SMY increased to 401 ± 22 and 278 ± 22 NmL/g VS, respectively, with an improved VS reduction of 47.8 ± 3.5 %. The pH dropped to 7.57 ± 0.14 with the VFA/ALK ratio of 0.309 ± 0.046 g HAc/g $CaCO_3$, indicating increased acidification due to the high biodegradability of FW. However, a steady balance between acidification and buffering was maintained (Fig. 1**b-d**). Compared to SS mono-digestion, AcoD significantly improved SBY, SMY, and VS reduction by 54.2 %, 46.3 %, and 23.2 %, respectively. Meanwhile, the digester fed with FW

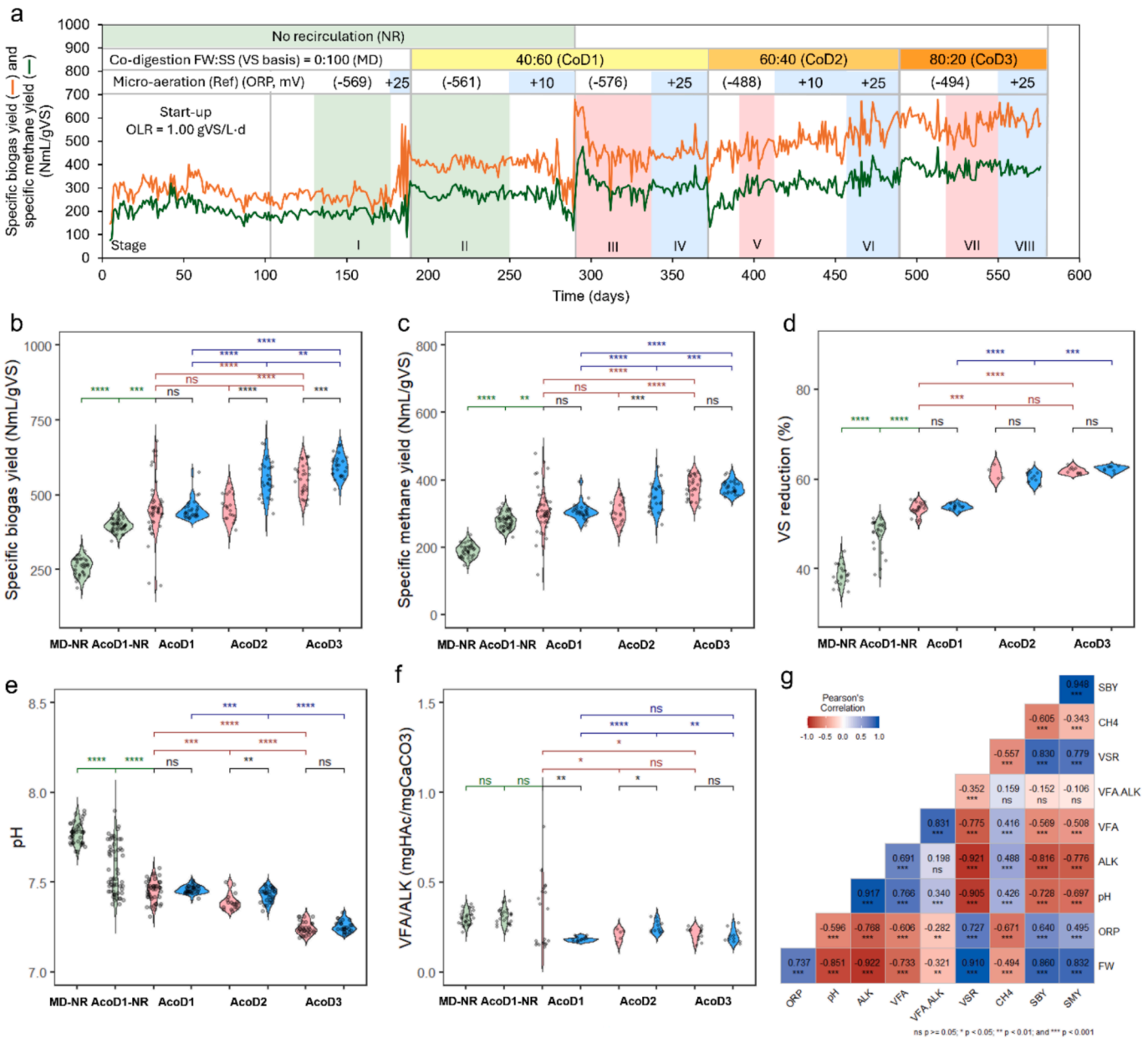


Fig. 1. Performance of continuous stirred-tank reactor at an organic loading rate of 2.25 g VS/L·d under anaerobic (pink) and micro-aerobic (blue) conditions, with varying co-substrate ratios of food waste (FW) to sewage sludge (SS) (FW:SS) (mono-digestion (MD) = 0:100, AcoD1 = 40:60, AcoD2 = 60:40, and AcoD3 = 80:20, VS basis) across stages I –VIII (a). Parameters include specific biogas yield (b), specific methane yield (c), VS reduction (d), pH (e), and VFA/ALK ratio (f), with their correlations shown in g. The data summary is available in the supplementary materials. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

as a mono-substrate at an OLR of 1.00 g VS/L·d failed after 10 days of operation due to VFA accumulation, with pH dropping to 6.80 and the VFA/ALK ratio reaching 0.530 g HAc/g $CaCO_3$, attributed to the rapid hydrolysis of FW. The stable operation of AcoD system with enhanced methane production was due to the synergistic effects, where SS provided alkalinity to the system, while FW contributed readily biodegradable organic matter (Chuenchart et al., 2024). Despite applying low-rate micro-aeration (+10 mV above the baseline ORP) during AcoD of FW and SS at the FW:SS ratio of 40:60 and an OLR of 2.25 g VS/L·d (Stage II), a significant portion of oxygen escaped into the headspace, leading to system instability. To address this issue, biogas was recirculated from the bottom of the digester starting on day 289 (Stage III) to enhance better oxygen mass transfer. Biogas recirculation ameliorated micro-aeration control, thereby improving the system performance and stability (Fig. 1**b-f**).

3.1.2. *Effects of co-substrate ratio on system performance and stability*

After successfully developing micro-aeration coupled with biogas recirculation, AcoD at the FW:SS ratio of 40:60 was evaluated under biogas recirculation (Stage III). Compared to without biogas recirculation, biogas recirculation significantly improved SBY, SMY, and VS reduction by 13.5 %, 10.8 %, and 11.9 %, respectively, yielding respective values of 455 ± 94 NmL/g VS, 308 ± 65 NmL/g VS, and 53.5 ± 1.4 %. Additionally, pH and the VFA/ALK ratio remained within the acceptable range at 7.45 ± 0.06 and 0.314 ± 0.188 g HAc/g $CaCO_3$, respectively. The significant improvement in performance after biogas recirculation was likely due to better mass transfer and solid–liquid interface exchange, as well as a shift in the microbial community (Zhao et al., 2020).

Under anaerobic conditions (Stages III, V, and VII), increasing the FW content in the co-substrate mixture to FW:SS ratios of 60:40 and

80:20 resulted in an SBY increase to 459 ± 43 and 552 ± 50 NmL/g VS, respectively. The corresponding SMY values were 305 ± 31 and 378 ± 32 NmL/g VS, respectively. Overall, the improvements in SBY and SMY were more pronounced at the FW:SS ratio of 80:20 compared to 40:60, which was attributed to the increased FW content (Karki et al., 2022). Additionally, for FW:SS ratios of 60:40 and 80:20, the pH remained stable at 7.39 ± 0.05 and 7.25 ± 0.04, respectively, with corresponding VFA/ALK ratios of 0.207 ± 0.033 and 0.214 ± 0.034 g HAc/g $CaCO_3$.

In contrast, consistent increases in SBY and SMY were observed under micro-aerobic conditions (Stages IV, VI, and VIII). Compared to the FW:SS ratio of 40:60, ratios of 60:40 and 80:20 improved SBY by 23.6 % and 32.2 %, resulting in 556 ± 59 and 595 ± 37 NmL/g VS, respectively. The corresponding SMY values also improved by 12.7 % and 23.2 %, reaching 345 ± 39 and 377 ± 19 NmL/g VS, respectively. Additionally, VS reduction increased from 53.8 ± 0.6 % to 60.3 ± 1.2 % and 62.3 ± 0.6 %, representing 12.1 % and 15.8 % increases, respectively, indicating higher conversion of VS into biogas. The rapid degradability of FW, which had 510 % more sCOD and 35 % more total carbon content than SS (see supplementary materials), likely contributed to the efficient conversion of VS in FW into biogas.

Overall, the higher FW content in co-substrate mixture led to increased SBY, SMY, and VS reduction, as evidenced by strong positive correlations (R > 0.70) (Fig. 1**.g**). Additionally, negative correlations between FW content and pH, ALK and VFA indicated rapid conversion of organic matter to VFA and biogas at higher FW levels in the co-substrate mixture.

3.1.3. Effects of micro-aeration on system performance and stability

The effects of micro-aeration on system performance varied with different FW and SS mixing ratios (Stage III-VIII), influenced by their aggregate characteristics on VFA production and consumption. During the initial phase (FW:SS ratio of 40:60, Stage IV), micro-aeration did not significantly impact the AcoD performance and stability parameters (SBY, SMY, VS reduction, and pH). The significant decrease in the VFA/ALK ratio from 0.556 to 0.150 g HAc/g $CaCO_3$ was due to the adaptation to biogas recirculation (Fig. 1**f**). At FW:SS ratio of 60:40, micro-aeration significantly improved SBY by 21.1 % (556 ± 59 NmL/g VS) and SMY by 13.1 % (345 ± 39 NmL/g VS) compared to anaerobic conditions (459 ± 43 and 305 ± 31 NmL/g VS, respectively). This performance improvement, despite no change in VS reduction, was likely due to the 35 % higher organic carbon content in FW than SS, leading to increased biogas production per unit of VS consumed. Additionally, micro-aeration had a pronounced effect on stability parameters, improving pH from 7.39 ± 0.05 to 7.43 ± 0.05 and the VFA/ALK ratio from 0.207 ± 0.033 to 0.254 ± 0.033 g HAc/g $CaCO_3$. At the FW:SS ratio of 80:20, only SBY increased by 7.8 % (from 552 ± 50 to 595 ± 37 NmL/g VS), while SMY, VS reduction, pH, and the VFA/ALK ratio remained nearly unchanged compared to anaerobic conditions.

Overall, higher FW content (i.e., FW:SS ratios of 60:40 and 80:20) enhanced SBY (CH_4, CO_2, and H_2) due to enhanced anaerobic fermentation of readily hydrolyzable and/or soluble organic matter within FW. The enhanced facultative activity under micro-aerobic condition likely have contributed to the increased CO_2 production. Consequently, increased availability of CO_2 can improve hydrogenotrophic methanogenesis, depending on the availability of hydrogen and substrate types (Nguyen et al., 2019). Despite variations in FW:SS ratios (from 40:60 to 80:20), pH and the VFA/ALK ratio remained fairly stable within the ranges of 7.20–7.60 and 0.180–0.320 g HAc/g $CaCO_3$, respectively, indicating a stable AD system. Furthermore, the concentrations of individual VFAs remained fairly low, with acetic acid below 400 mg/L and all other VFAs below 100 mg/L. The effects of micro-aeration were inconsistent across different co-substrate ratios, likely due to the low OLR. Therefore, further experiments were conducted to investigate its impact on digester stability at higher OLRs.

3.2. Effects of organic loading rate and micro-aeration on system performance and stability

3.2.1. Effects of organic loading rate on system performance and stability

The FW:SS ratio of 80:20, which yielded the highest SBY and SMY at an OLR of 2.25 VS/L·d, was selected to investigate the effect of micro-aeration on digester stability at higher OLRs of 4.0 and 6.0 g VS/L·d. To improve the reproducibility, second and third trials were conducted in a parallel-operated second digester (Fig. 2**a**). Overall, the results were highly consistent across the trials, except during the failure stage at an OLR of 6.0 g VS/L·d, where data were excluded from analysis.

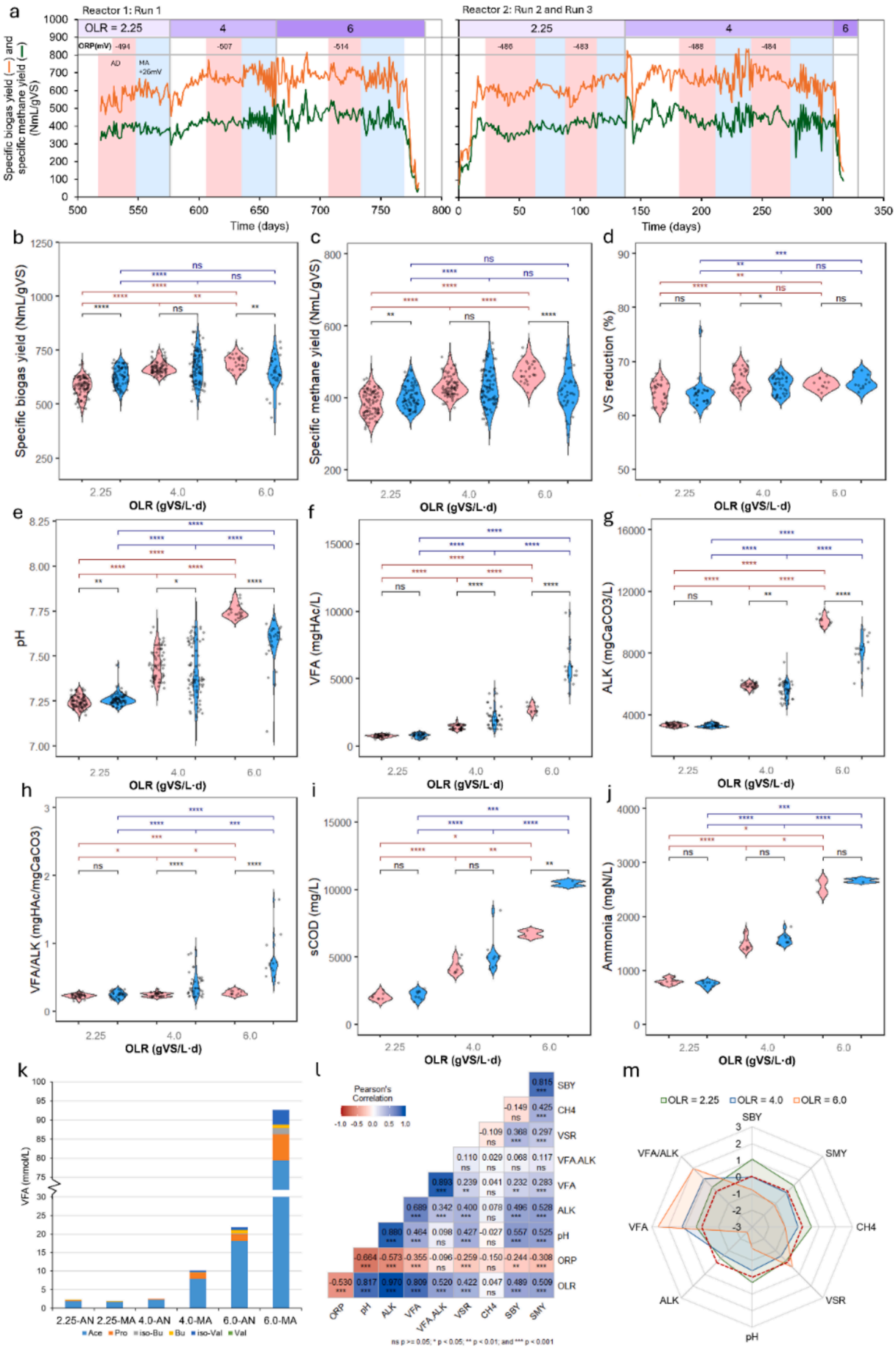
Under anaerobic conditions, at OLRs of 2.25, 4.0, and 6.0 g VS/L·d, SBY values were 578 ± 46, 669 ± 29, and 693 ± 32 NmL/g VS, respectively, with corresponding SMY values of 385 ± 35, 434 ± 31, 470 ± 29 NmL/g VS. Increases in OLRs from 2.25 to 4.0 and 6.0 g VS/L·d corresponded to improvement in SBY by 15.7 % and 19.7 %, and SMY by 12.7 % and 22.0 %, respectively. VS reduction showed a similar trend, reaching 64.2 ± 1.9 %, 66.7 ± 2.0 %, and 65.8 ± 1.0 % at OLRs of 2.25, 4.0 and 6.0 g VS/L·d, respectively, representing increases of 3.8 % and 2.5 % compared to the lowest OLR tested. The improvements in performance parameters (SBY, SMY, and VS reduction) at an OLR of 4.0 g VS/L·d, followed by decreases or insignificant changes at an OLR of 6.0 g VS/L·d suggest potential system inhibition at the highest OLR tested (Fig. 2 **a-c**).

Under micro-aerobic conditions, increasing OLR from 2.25 to 4.0 g VS/L·d resulted in an increase in SBY from 626 ± 44 to 669 ± 76 NmL/g VS and SMY from 403 ± 33 to 430 ± 53 NmL/g VS, reflecting improvements of 6.9 % and 6.7 %, respectively. Although the pH increased from 7.26 ± 0.03 to 7.41 ± 0.15, the VFA/ALK ratio also increased from 0.243 ± 0.048 to 0.387 ± 0.175 g HAc/g $CaCO_3$. Additionally, total VFA concentration significantly increased from 1.9 to 10.2 mmol/L (5.4-fold increase), indicating initial VFA accumulation during micro-aeration (Fig. 2**k**). However, at an OLR of 6.0 g VS/L·d, SBY and SMY decreased to 646 ± 75 and 419 ± 53 NmL/g VS, respectively, prior to system failure due to excessive VFA accumulation (Fig. 2**d**).

As OLR increased, VFA, ALK, total ammonia, and sCOD concentrations consistently and significantly increased in both conditions, as indicated by moderate to strong correlations (Fig. 2**f-g, i-j, and l**). This is likely due to a high ammonia and organic nitrogen content in both FW and SS, as reflected by the low C/N ratio of 11.9 at the FW:SS ratio of 80:20. Total ammonia levels increased to 1500 ± 120 and 2560 ± 160 mg N/L but remained below the inhibition threshold for AD (i.e., 4000 mg N/L) (Yan et al., 2020). However, this increase in total ammonia could partially inhibit acetoclastic methanogenesis, with an IC_{50} of 2130 mg N/L (Wang et al., 2022). The system failure at an OLR of 6.0 g VS/L·d suggests that the optimal OLR for FW and SS ACoD at the FW:SS ratio of 80:20 (VS basis) falls below 6.0 g VS/L·d for both micro-aerobic and anaerobic conditions.

3.2.2. Effects of micro-aeration on system performance and stability

The effects of micro-aeration on SBY and SMY varied across OLRs, demonstrating enhancement, neutrality, or reduction (Fig. 2**b, c, and m**). This was reflected in the SMD (negative, zero, and positive) and SPI values of 0.483, −0.213, and −0.493 for OLRs of 2.25, 4.0 and 6.0 g VS/L·d, respectively (Fig. 2**m**). At an OLR of 2.25 g VS/L·d, micro-aeration significantly improved SBY and SMY by 8.3 % (from 578 ± 46 to 626 ± 44 NmL/g VS) and 4.4 % (from 385 ± 35 to 403 ± 33 NmL/g VS), respectively. Additionally, pH and the VFA/ALK ratio remained fairly stable at 7.25 ± 0.03 and 0.228 ± 0.030 g HAc/g $CaCO_3$, respectively. However, at an OLR of 4.0 g VS/L·d, micro-aeration had no significant effect on either SBY or SMY. During micro-aeration, VS reduction, pH, ALK remained unchanged at 65.7 ± 1.6 %, 7.41 ± 0.15, and 5619 ± 551 mg $CaCO_3$/L, respectively. However, increases in VFA from 1434 ± 179 to 2109 ± 742 mg HAc/L and the VFA/ALK ratio from 0.244 ± 0.028 to 0.389 ± 0.175 g HAc/g $CaCO_3$ indicated the beginning of VFA accumulation during micro-aeration.



(caption on next page)

Fig. 2. Performance of anaerobic (pink) and micro-aerobic (light blue) conditions during FW and SS mixing ratio of 80:20 (i.e., FW:SS = 80:40, VS basis) and varying organic loading rates (OLRs) of 2.25, 4.0, and 6.0 g VS/L·d (a). Parameters include specific biogas yield (b), specific methane yield (c), VS reduction (d), pH (e), total volatile fatty acid (VFA) (f), alkalinity (ALK) (g), VFA/ALK (h), sCOD (i), total ammonia (j), and individual VFA (k). Experiments were conducted in triplicate for more than one HRT (20 days) after reaching steady ALK. For the OLR of 6.0 g VS/L·d, sCOD and total ammonia data were not replicated due to system failure in trials 2 and 3. Effects of OLR and micro-aeration were analyzed using Pearson's correlation (l) and radar plot of standard mean differences (SMD) (m). A data summary is available in the supplementary materials. Asterisks (*) indicate a significant difference by *t*-test ($p < 0.05$:*, $p < 0.01$:**, $p < 0.01$:***). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

At an OLR of 6.0 g VS/L·d, micro-aeration significantly reduced SBY and SMY by 6.7 % (from 693 ± 32 to 646 ± 75 NmL/g VS) and 10.9 % (from 470 ± 29 to 419 ± 53 NmL/g VS), respectively. Additionally, pH dropped from 7.75 ± 0.04 to 7.57 ± 0.13 and ALK decreased from 10100 ± 300 to 8160 ± 930 mg $CaCO_3$/L. Furthermore, the increase in VFA levels from 2760 ± 310 to 5930 ± 1520 mg HAc/L and in the VFA/ALK ratio from 0.274 ± 0.034 to 0.757 ± 0.314 g HAc/g $CaCO_3$ indicated inhibition due to VFA accumulation during micro-aeration. Acetic acid concentrations increased significantly, reaching 4690–4860 mg HAc/L (78.1–80.8 mmol/L), while propionic acid levels increased to 426–549 mg HPro/L (5.8–7.8 mmol/L) (Fig. 2**k**). Furthermore, titrated total VFAs (~10,000 mg HAc/L) exceeded the sum of identified individual VFAs (~5500 mg HAc/L), suggesting the accumulation of other organic acids, such as medium- and long-chain fatty acids. The increase in sCOD from 6700 to 10,400 mg COD/L after micro-aeration further indicated the potential accumulation of soluble organic compounds. However, micro-aeration did not affect ammonia concentration.

SMD and SPI also indicated that micro-aeration led to lower pH and ALK, along with higher VFA levels and VFA/ALK ratios, suggesting VFA accumulation and inhibition of methanogenesis (Fig. 2**m**). At an OLR of 4.0 g VS/L·d, increased hydrolysis and acetogenesis were observed, as indicated by higher total VFA and higher individual VFA levels (Fig. 2**d, f, and k**). Therefore, no significant differences in SBY and SMY were observed between the micro-aerobic and control conditions at this OLR, while system inhibition occurred at an OLR of 6.0 g VS/L·d.

Despite the previous study reporting the effectiveness of micro-aeration in overcoming VFA accumulation during the AD of lignocellulosic feedstock, such as Napier grass (*Pennisetum purpureum*) at an OLR of 5.0 g VS/L·d (Nguyen et al., 2019), this study observed inconsistent effects of micro-aeration on AcoD. This inconsistency can be attributed to the highly alkaline nature of SS, the readily hydrolyzable characteristics of FW, and the inherent competition between homoacetogens and hydrogenotrophic methanogens, which is influenced by the nature of co-substrates. When ALK is below 5,000 mg $CaCO_3$/L, micro-aeration can effectively control accumulated VFAs by enhancing VFA oxidation. However, when ALK exceeds this threshold, micro-aeration may deteriorate digester performance due to increased VFA accumulation. Additionally, during the AD of readily hydrolyzable substrates at high OLRs, micro-aeration may exacerbate VFA accumulation as facultative bacteria ferment more organic matter to organic acids, disrupting the VFA-ALK balance.

In systems where hydrolysis is the rate-limiting step—such as those digesting lignocellulosic biomass and animal manure (e.g., corn straw (Fu et al., 2016; Zhu et al., 2022), Napier grass (Nguyen et al., 2019), poultry litter and wheat straw (Zhan et al., 2023))—micro-aeration has been shown to improve AD performance by promoting the activity of hydrolytic bacteria (Fu et al., 2023). In contrast, during the AD of readily hydrolyzable substrates, such as FW (Xu et al., 2014) and wastewater (Salomé Duarte et al., 2024), micro-aeration can lead to the accumulation of sCOD and organic acids, especially at high OLRs.

3.3. Microbial community dynamics

3.3.1. Bacterial communities

Amplicon sequencing of the 16S rRNA gene revealed that varied co-digestion and OLR periods greatly shifted community structure, especially with respect to the initial mono-digestion phase. A non-metric multidimensional scaling (NMDS) plot based on Bray-Curtis distance matrix indicated that the divergence of the microbial community from mono-digestion to FW:SS mixing ratios resulted in higher partitioning of microbial groups along NMDS 2 axis, suggesting an influence of the feeding regime on community structure (AMOVA, p-value < 0.05). Although it was apparent that the communities at an OLR of 6.0 g VS/L·d were relatively similar, the shift along both NMDS 1 and NMDS 2 axes was significant compared to the rest of the samples, in addition to the separation of the microbial groups at lower OLRs (HOMOVA, p-value < 0.05) (see supplementary materials).

A calculation of diversity metrics included the Shannon Index (H'), richness (S), and invSimpson (1/D). Results showed an initial increase in S during co-digestion and at different FW:SS ratios, but a general decreasing trend in diversity was observed across biomass samples at increasing OLRs (see supplementary materials). The mono-digestion period, indeed, had the highest overall diversity and eventually decreased from 3.51 to 2.02 for H' and from 223 to 145 for S. Evaluating the relative abundance (RA) of populations throughout the experiment revealed a prominent shift in the three most abundant phyla, Chloroflexi, Bacteroidetes, and Firmicutes, known to be either strict or facultative anaerobic fermentative groups. The mono-digestion period was dominated by Firmicutes (with RA 46.0 %), while Chloroflexi thrived at increased FW:SS ratios at an OLR of 2.25 g VS/L·d, with RA ranges between 40–50 %, after which Bacteroidetes were observed to dominate when OLR was increased to 4.0 g VS/L·d and then to 6.0 g VS/L·d, reaching a maximum of 68.7 % (Fig. 3**a**). It is also noteworthy to mention that Firmicutes and Bacteroidetes abundances were relatively stable during the increased FW:SS ratios. These groups were previously found to correlate with VFA production, namely acetic and propionic acids, during the long-term mesophilic AD of FW and SS at increasing mixing ratios (Owusu-Agyeman et al., 2023). A closer look at the genus-level RA distribution revealed top genera that likely contributed to the shift in microbial communities (Fig. 3**b**). Genera that contributed to significant changes during the co-substrate variation period were unclassified Ruminococcaceae (homo-acetogens), unclassified Syntrophomonadaceae (anaerobic syntrophs), and unclassified Actinomycetaceae (facultatively anaerobic bacteria), with increasing RAs in general, and unclassified Pseudomonadaceae with decreasing overall RAs (*t-test*, *p*-value < 0.05). Based on these observations, it is apparent that the microbial groups varied greatly during the co-substrate variation period, mainly because of adaptation to the complex substrate being introduced throughout the increasing FW:SS ratios and micro-aeration application. On the other hand, the genera that contributed to significant changes during the OLR variation period were unclassified Bacteroidetes (increased RAs in general), *Syntrophomonas*, unclassified Bacteroidales, *Bacteroides*, unclassified Syntrophaceae, and unclassified Methanobacteriaceae, with decreasing RAs in general (*t-test*, *p*-value < 0.05). Genera that significantly changed during both periods were related to unclassified *Clostridiales* spp., unclassified *Anaerolineaceae* spp., unclassified *Dysgonomonadaceae* spp., *Aminipila*, unclassified *Prolixibacteraceae* spp., *Sedimentibacter*, *Proteiniborus*, and *Romboutsia*. For example, unclassified *Anaerolineaceae* significantly increased from 12.6 % to 47.3 % initially and decreased to 3.30 % at the end of the experiment, while *Aminipila* decreased overall (from 2.14 % to 0.06 %), while *Sedimentibacter* initially decreased from 0.72 % to 0.06 % and then increased again to 0.10 %, respectively (*t-test*, *p*-value < 0.05). These genera are either strict anaerobes or facultative anaerobes, which indicates that the impact of micro-aeration on community structure should not be overlooked even at low RA. It is likely that micro-aeration

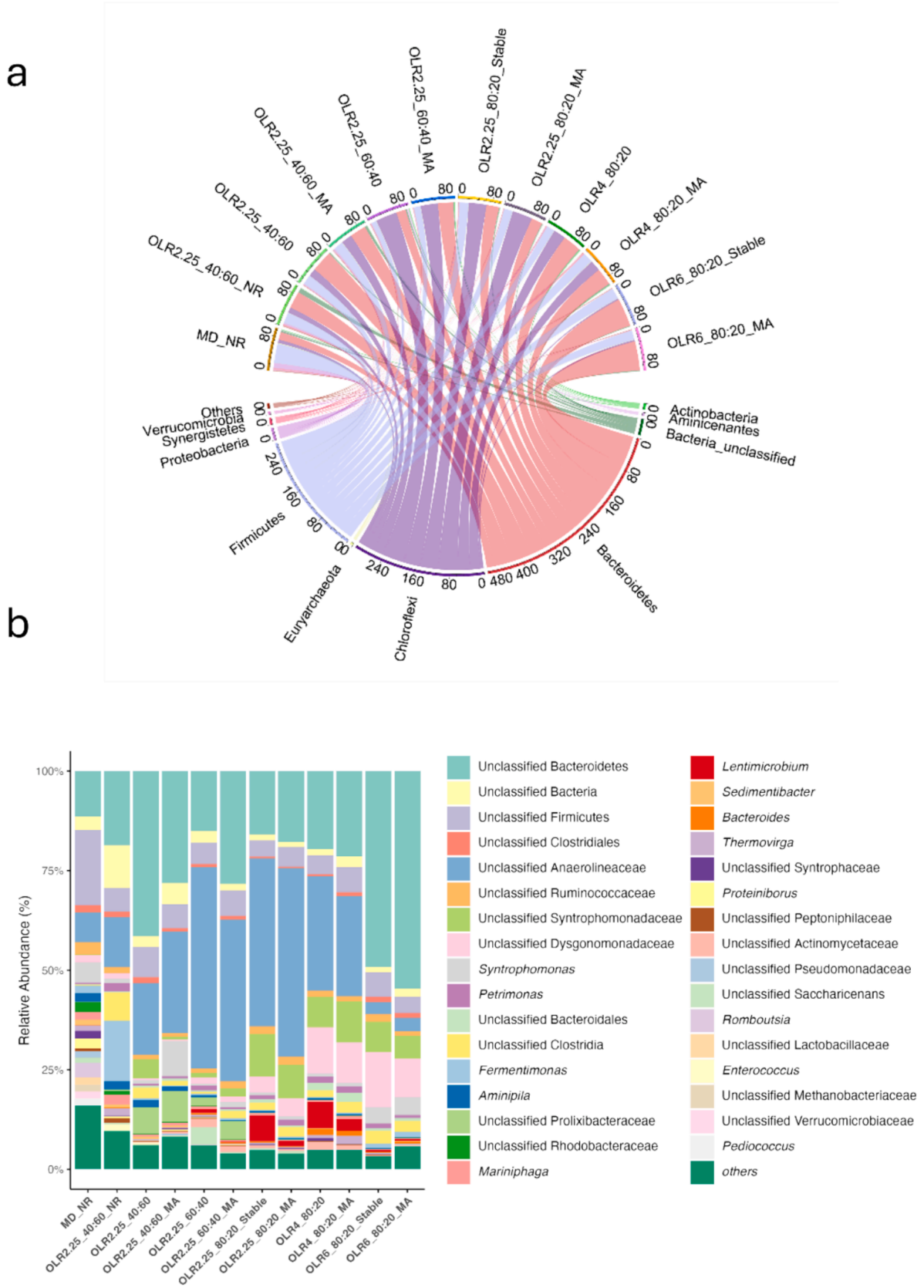


Fig. 3. Chord diagram showing the relative abundance (RA in %) distribution of the top 10 phyla of the microbial community in the samples taken during the experiment. The chord colors refer to each phylum plotted and the microbial community axis labels are cumulative relative to the phylum in each sample (a). Normalized stacked bar chart showing the relative abundance (RA in %) distribution of the genera with RA > 1 % in at least one sample from the samples taken during the experiment (b). NR: Non recirculation and OLR: Organic loading rate.

conditions did help fermentation of more complex substrates, as demonstrated by the results during the increased FW:SS ratios. Since genera often comprise a broad selection of strict anaerobes and facultative groups, an identification at species level with a minimum 90 % confidence level revealed the presence of facultative anaerobes, such as *Actinomyces provencensis* (97.6 % confidence), *Rhodobacter xinxiangensis* (98.8 % confidence), *Mariniphaga sediminis* (91.7 % confidence), and *Mariniphaga anaerophila* (93.3 % confidence). The presence of facultative groups did not seem to alter the system's performance, yet no strong conclusion could be drawn from the current experiment. Further, these observations are in accordance with the stable performance recorded at an OLR of 4.0 g VS/L·d, suggesting key microbial groups' abilities to thrive under micro-aerobic conditions.

3.3.2. Archaeal communities

Archaeal community structure changed considerably during co-digestion and OLR variation periods compared to the mono-digestion phase. The archaeal community was dominated by three main families: *Methanobacteriaceae* spp. initially during the mono-digestion phase, *Methanotrichaceae* spp. starting co-digestion with consistent high abundances, and *Methanospirillaceae* spp. at higher OLRs, with gradually increased abundances at OLR of 4.0 g VS/L·d and OLR of 6.0 g VS/L·d, respectively, which were aligned with the increased SBY and SMY during these phases (Fig. 4**a**). Among the genera classified, *Methanobacterium* spp., *Methanobrevibacter* spp., *Methanoculleus* spp., *Methanolinea* spp., *Methanomethylovorans* spp., and *Methanosarcina* spp. were only observed during the co-digestion period at an OLR of 2.25 g VS/L·d, after which a higher selectivity occurred at OLRs of 4.0 and 6.0 g VS/L·d. More specifically, three genera related to *Methanothrix* spp. (*Methanothrix soehngenii* sequence identified at 99.2 %), *Methanospirillum* spp. (*Methanospirillum hungatei* sequence identified at 95.3 %), and *Methanomassiliicoccus* spp. contributed to more than 90 % of the archaeal community at the end of the experiment, with the latter two having slightly increased RAs (Fig. 4**b**). These observations indicated that although methanogens were resilient to the different conditions, given that CH_4 production was not affected initially, an OLR of 6.0 g VS/L·d likely served as a threshold for these selective methanogens to survive. Predictive functionality capabilities of the microbial community were further studied using Tax4FUN2 and KEGG. Key modules and corresponding unique orthologs that relate to the methanogenesis were selected to identify predicted genes encoding for key enzymes (see supplementary materials), namely acetoclastic methanogenesis (M00357), hydrogenotrophic methanogenesis (M00567), and the reductive acetyl-CoA pathway (M00377) for homo-acetogens (Rama et al., 2024). Upon starting co-digestion, hydrogenotrophic methanogenesis and homoacetogenesis capabilities increased (a shift from acetoclastic methanogenesis), which partially explains the positive effect on the system's performance. Interestingly, all three were observed to have decreased capabilities at the end of the experiment (Fig. 5**a**). Further, hydrogenotrophic methanogenesis increased when biogas recirculation was introduced, which aligns with the positive effects reported on system's stability, likely due to the readily available CO_2. In addition, the decrease in acetoclastic pathway and *Methanothrix* spp. at an OLR of 6.0 g VS/L·d was likely linked to the decreased SMY observed. *Methanospirillum* spp. and *Methanothrix* spp. are both capable of using CO_2 or acetate to produce CH_4, while *Methanomassiliicoccus* spp. can use methylamine/dimethylamine/trimethylamine to produce CH_4. On the other hand, the dominance of *Methanospirillum* spp. is explained by their ability to survive under oxygenated environments due to their antioxidant features (Lyu and Lu, 2018). Species related to *Methanospirillum,* namely *M. hungatei*, can produce CH_4 by utilizing H_2 and formate as electron donors to reduce CO_2 in syntrophy with *Syntrophomonas*. Coincidentally, *Syntrophomonas* genera identified in our study significantly increased during the highest OLR tested (i.e., 6.0 g VS/L·d) (*t*-test, p-value < 0.05) with two species identified: *Syntrophomonas bryantii* (at confidence level 97.2 %) and *Syntrophomonas wolfei* (at confidence level 94.9 %). These two species are phylogenetically closely related and have been studied for cooperation via butyrate oxidation (Crable et al., 2016; Walker et al., 2019). In addition, microbial communities related to homoacetogenesis and hydrogenotrophic methanogenesis pathways have been reported to compete for H_2, and the higher capabilities of key enzymes with the homoacetogenesis pathway likely relate to the acetate accumulation observed during the highest OLR, especially that unclassified *Ruminococcaceae,* among key homoacetogens*,* were also abundant (Karekar et al., 2022). Homoacetogens have higher adaptability to acidic and alkaline conditions, at which they outcompete hydrogenotrophs as well (Mosbæk et al., 2016; Pan et al., 2021). Further, an alternative acetate-degrading community involves a synergy between syntrophic acetate oxidizers (SAOB), slow growers, and hydrogenotrophic methanogens. A few syntrophic acetate oxidizing genera were merely present in the microbiome consortia, with decreased RAs at < 1 % (see supplementary materials), indicating that this pathway did not likely prevail under the reported conditions. Overall, methanogens were resilient to increased FW:SS ratios and micro-aeration, while diverse pathways were maintained at an OLR of 4.0 g VS/L·d.

3.3.3. Effects of operational parameters on microbial communities

The potential relationship between the experimental parameters and the abundance of microbial communities was further studied using the RDA and the significance was tested using ANOVA (Fig. 5**b**). Only the most abundant phyla were plotted to avoid cluttering. The experimental parameters chosen included the different OLRs, FW:SS ratios, and SPI. OLR and FW:SS ratio constraints significantly affected the microbial composition (ANOVA, p-value < 0.05). In addition, the two RDA components were found to explain 82.3 % of data variability, and that illustrated the associations between the variables and experimental conditions. The length of the arrow indicates the extent to which the selected environmental variables explain the taxonomy, while arrows in opposite directions show a negative correlation. The results showed a strong positive correlation between OLR and Bacteroidetes and a weaker correlation with OLR and Spirochaetes. SPI, however, showed the highest positive correlation with Aminicenantes, an organotroph with carbohydrates and proteinaceous fermentation metabolism (Kadnikov et al., 2019), and a weaker correlation with Actinobacteria. Further, Chloroflexi correlated the most with FW:SS ratio of 60:40, as similarly indicated above.

Nevertheless, micro-aeration did not seem to alter microorganisms while additional CO_2 was supplied and was not responsible for the system failure. Despite no strong evidence in the current results to explain sCOD accumulation and hydrolysis inhibition, further investigation of microbial activities needs to be conducted to understand the prevailing mechanisms.

4. Proposed mechanisms

The positive effects of micro-aeration on system performance and stability were observed when ALK was lower than 5000 mg $CaCO_3$/L, under conditions where OLR was 2.25 g VS/L·d, and FW:SS ratios were 60:40 (trial 1) and 80:20 (trials 1–3). However, these positive effects became insignificant when ALK exceeded 5000 mg $CaCO_3$/L under conditions, including 1) OLR of 2.25 g VS/L·d with the FW:SS ratio of 40:60 and 2) OLRs of 4.0 and 6.0 g VS/L·d with the FW:SS ratio of 80:20. At lower ALK, VFAs can be converted to CO_2 during micro-aeration, which can then be utilized by homoacetogens to produce acetate, which is subsequently converted into CH_4 via acetoclastic methanogenesis (Fig. 5**c**). Under these conditions, hydrolysis becomes a rate-limiting step due to the low concentrations of metabolic intermediates (e.g., acetate and CO_2). Therefore, micro-aeration enhances hydrolysis and acidogenesis by promoting the growth and activity of diverse facultative anaerobes.

In contrast, when ALK is higher than 5000 mg $CaCO_3$/L, owing to high OLRs and conversion of substrates, including complex biopolymers,

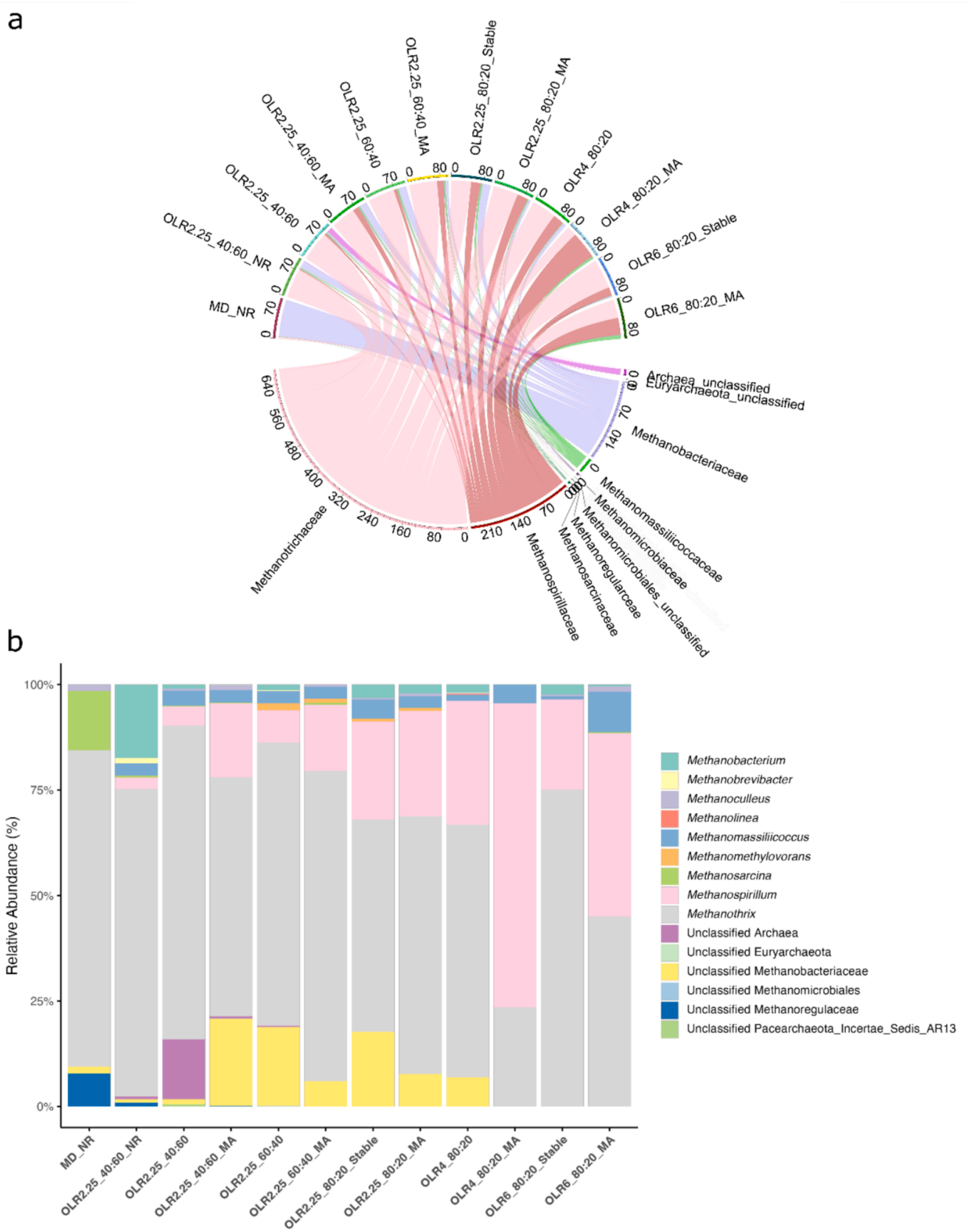


Fig. 4. Chord diagram showing the relative abundance (RA in %) distribution of the archaea phyla of the microbial community in the samples taken during the experiment. The overlapping labels in order are: Methanomicrobiacaea, Methanomicrobiales_unclassified, Methanoregulacaea, and Methanosarcinacaea. The chord colors refer to each phylum plotted. The values for the archaeal community are cumulative. (a). Normalized stacked bar chart showing the relative abundance (RA in %) distribution of archaea genera of the microbial community in the samples taken during the experiment. The archaea data was selected, and RA were calculated (b). NR: Non-recirculation and OLR: Organic loading rate.

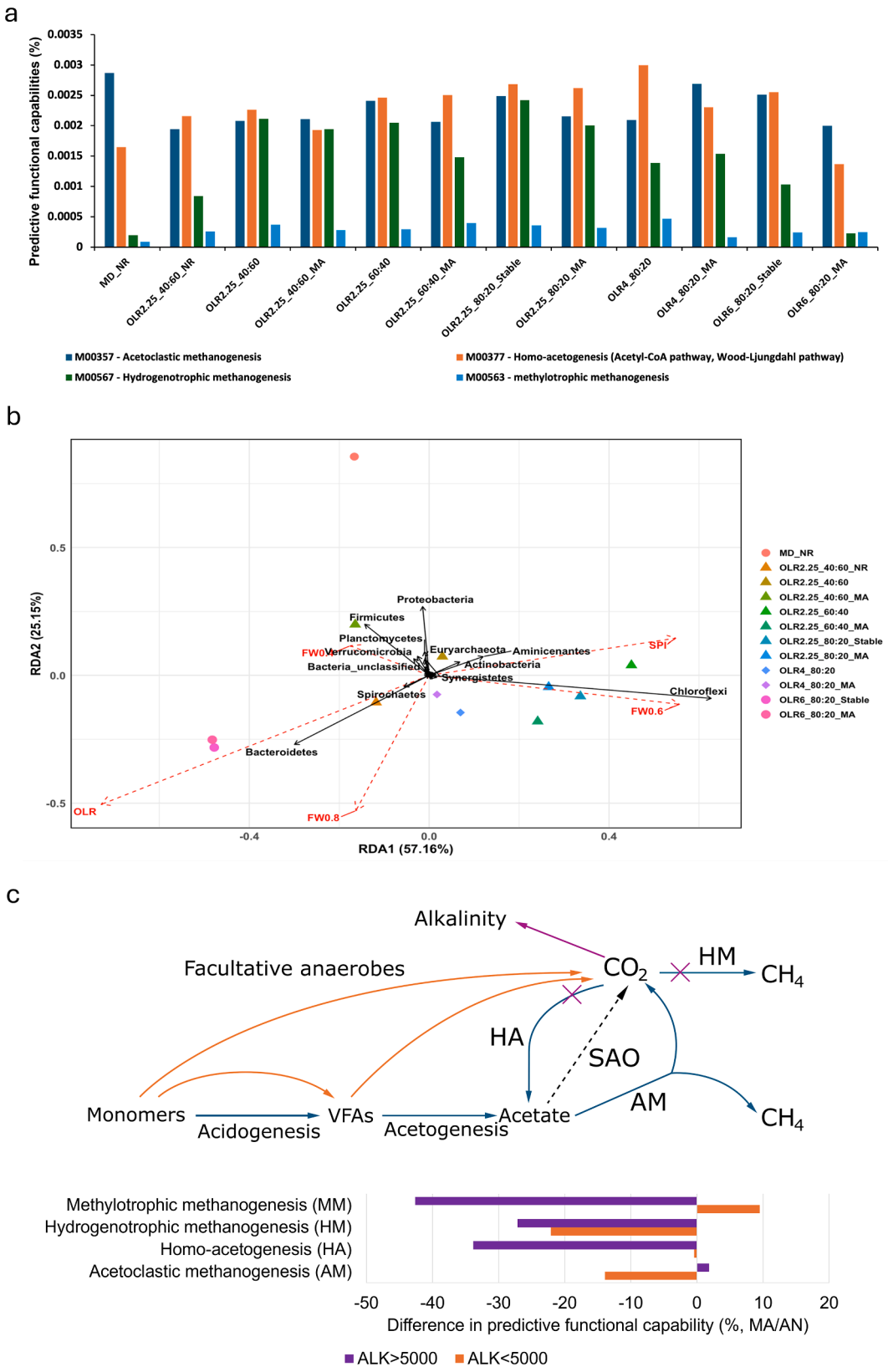


Fig. 5. Bar chart showing the predictive functional capabilities (in %) distribution of the microbial communities in methane metabolism: acetoclastic (acetate to methane), hydrogenotrophic (CO_2 to methane), methylotrophic (methanol/ methylamine/dimethylamine/trimethylamine to methane), homoacetogenesis (via the acetyl-CoA pathway). The prediction was done using TAX4FUN2 and KEGG pathways modules (a). Redundancy analysis results chart. The top abundant phyla are shown in black arrows. The environmental conditions are in dashed red lines (b). Proposed mechanisms of micro-aeration under the influence of alkalinity in food waste and sewage sludge co-digestion with percent different predictive functional capabilities of micro-aerobic condition, compared to anaerobic condition (c). NR: Non-recirculation; OLR: Organic loading rate; SPI: Standardized performance index. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

proteins, amino acids, and polysaccharides, the partial pressure of CO_2 decreases. Methanogenesis becomes a rate-limiting step due to the accumulation of intermediate metabolites. The decrease in CO_2 partial pressure, which is highly influenced by carbonaceous ALK, could inhibit hydrogenotrophic methanogenesis (Mayumi et al., 2013). In this study, at OLRs of 4.0 and 6.0 g VS/L·d, micro-aeration significantly inhibited hydrogenotrophic methanogenesis, resulting in approximately 4.1-fold increase in total VFAs concentration. Therefore, micro-aeration did not improve system performance under OLRs of 4.0 and 6.0 g VS/L·d, with ALK exceeding 5000 mg $CaCO_3$/L. Moreover, at high OLRs, where methanogenesis was inhibited due to system acidification (Acetic acid = 1090 ± 330 mg/L), micro-aeration further impaired both methanogenic pathways.

Several studies have reported that micro-aeration enhances performance of AD under low ALK conditions, such as during the digestion of Napier grass (ALK = 2880–3590 mg $CaCO_3$/L) (Nguyen et al., 2019), poultry litter and wheat straw (ALK = 1240–1550 mg $CaCO_3$/L) (Zhan et al., 2023), swine wastewater (ALK = 590–640 mg $CaCO_3$/L) (Guimarães de Oliveira et al., 2021), and SS (ALK = 1065 mg $CaCO_3$/L) (Morello et al., 2021). Conversely, due to the presence of homoacetogenic anaerobes, micro-aeration led to acid accumulation during AD of organic fraction of municipal solid waste under high ALK conditions (ALK = 15480–18000 mg $CaCO_3$/L) (Bacab et al., 2020). While this trend is consistent across different substrates, the improvement in hydrolysis and low ALK of the (co-)substrates likely played a significant role in AD system performance and stability. Additionally, the potential contribution of micro-aeration to VFAs and medium- to long-chain fatty acids production should not be overlooked (Duarte et al., 2018; Fu et al., 2023; Xu et al., 2014).

5. Conclusions

Micro-aeration contributed to the AcoD system performance primarily through improved hydrolysis. Performance improvements were observed only in FW:SS ratios of 60:40 and 80:20 at an OLR of 2.25 g VS/L·d, when ALK was below 5000 mg $CaCO_3$/L. Both obligate and facultative anaerobes, including unclassified Ruminococcaceae, unclassified Syntrophomonadaceae, and unclassified Actinomycetaceae, were predominant during micro-aeration. The standardized performance index indicated Aminicenantes as the key bacteria responsible for improving system performance during micro-aeration. The nature of co-substrates and system alkalinity played an important role in the positive effects of micro-aeration, likely due to bicarbonate concentration, homoacetogenic activity, and the absence of syntrophic acetate oxidizers.

CRediT authorship contribution statement

Wachiranon Chuenchart: Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Christelle Sawaya:** Writing – original draft, Visualization, Investigation, Formal analysis. **K. C. Surendra:** Writing – review & editing, Supervision, Conceptualization. **Adam L. Smith:** Writing – review & editing, Supervision. **Samir Kumar Khanal:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors acknowledge funding from the Research Capacity Fund (Hatch: HAW 05050-H) and Multi-State Research (HAW 5042-R) through the United States Department of Agriculture (USDA), College of Tropical Agriculture and Human Resources at the University of Hawai'i at Mānoa (UHM). The authors would also like to acknowledge funding support from Biomass Energy Systems Inc. (BESI). Special thanks to Ty Shitanaka and Kyle Rafael Marcelino at UHM for their helpful discussion. Christelle Bou Nehme Sawaya is partially funded by the Environmental Research & Education Foundation, Raleigh, NC.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2025.132398.

Data availability

Data will be made available on request.

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